Online Appendix

The Political Class in Citizens' Minds: Mapping Public Perceptions of Politicians

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Appendix A – Sample Characteristics

Table A.1: Sample characteristics

Variable	% respondents	% quota
Gender		
Female	50.1	49.9
Male	49.3	50.1
Other/Do not want to answer	0.6	—
Age group		
18-34	31.5	31.2
35-49	25.2	25.2
50-64	27.6	27.6
65-75	15.7	16.0
Region of residence		
Capital region	32.9	32.6
Zealand	13.7	14.1
Southern Denmark	20.6	20.5
Central Jutland	22.9	22.8
Northen Jutland	9.8	9.9
Education		
Primary education	7.8	—
Upper secondary, general	12.1	—
Upper secondary, vocational	21.1	—
Short-cycle tertiary	9.4	—
Medium-cycletertiary education	30.8	—
Long-cycle tertiary (incl. PhD)	18.4	—
Other education	0.2	—
Personal income		
Less than 100.000 DKK	7.6	—
100.000 to 199.999 DKK	13.7	—
200.000 to 299.999 DKK	15.0	—
300.000 to 399.999 DKK	16.8	—
400.000 to 499.999 DKK	13.3	—
500.000 to 599.999 DKK	6.9	—
600.000 to 699.999 DKK	4.1	—
700.000 to 799.999 DKK	3.1	—
800.000 to 999.999 DKK	1.7	—
1.000.000 DKK or more	1.5	—
Don't know/refuse	16.3	—
Subjective class		
Working class	12.3	—
Lower middle class	12.5	—
Middle class	48.8	—
Upper or upper middle class	16.8	—
Don't know	5.8	—
Do not belong to any class	3.8	—

Note: N = 1302. Quota percentage are the actual share in the population at the time of the survey. These percentages are based on data from Danmarks Statistik.

Appendix B – Free Association Task

This section contains the appendices relevant to the analysis of the free association task.

Appendix B1 – Professions in the Free Association Task

Table B1.1 shows the different professions in the Free Word Association Test. The test consisted of three rounds. In round 1 and round 3, each respondents were randomly presented with one of the professions.

Table B1.1: Professions in the Rounds of the Association Test

Round 1	Round 2	Round 3
Doctors	Politicians	Journalists
Physiotherapists		Social workers
Engineers		Bud drivers
Librarians		Farmers
Midwives		Bank advisors
Teachers		Police officers
Priests		Lawyers
Building constructors		Nurses

Appendix B2 – List of Synonyms

Table B2.1 shows the synonyms used in the analysis. I have manually exported all associations to an Excel file, read through them, and created a list of synonyms.

Table B2.1: Synonyms in Free Word Association Task

Word	Synonym
akademiker	akademi*
debat/diskussion	debat*; diskussion*; disskussioner
egoistisk	egen vinding; egeninteresse; ego; egocentri*; egoer; egois*; egotrippere; egennyttige; egenrådig; selvcentre*; selvisk*; selviske; selvoptagethed; selvorienteret; tænker kun op sig selv; tænker kun på dem selv; tænker kun på sig selv; tænker på sig selv
elite	elitære
engagement	engagement; engageret
folketinget	folketinget; christiansborg; borgen; folketing
folkevalgt	folkevalgt*
høj_løn	højtbetal*; højtløn*; fed hyre; fede lønninger; meget i løn; meget løn; god løn; høje lønninger; høj indkomst; højtlønnede; overbetalt*; store lønninger; tjener godt; velløn*; god indkomst; løntung; tjener alt for meget; får for mange penge

Table B2.1: Synonyms in Free Word Association Task *(continued)*

Word	Synonym
højtuddannede	djøf'ere; høj uddannelse; boglige uddannet; boglig uddannelser; boglige; højere uddannelse; højtudda*; lang uddannelse; lang videregående uddannelse; statskundskab*; uddannet; universitetsuddannet; veluddannede; uddannede; djøfisk
korrupt	korrupte; korruption
levebrødspolitikere	karrierepolitiker*; levebrød*; karrieremennesk*
løgnere	løgnagtige; løgne*; løgnhals*; lyver
overklasse	overklass*; topklasse; topppen; toppen af samfund; højtstående
partier	parti; partier*
pensionsvilkår	pension*; ministerpension; livstidspension
posh	dresset; dyrt tøj; habitter; jakkesæt; pænt tøj; pænt klædt; slips; suit&tie; velklædte; jakkesæt, makeup og stiletter; klassisk kedeligt tøj
regering	regering; regeringen
rige	rig; rige
samfund	samfundet; samfund
upålidelige	upålidelig*
utroværdig	utroværdig*; ikke til at stole på
uærlige	uærlig*
virkelighedsfjern	knap så virkelighedsnære; mangler forståelse for virkeligheden; virkelighedsfje*; ude af trit med virkeligheden; langt fra virkeligheden; uden virkelighedssans; lever i en anden virkelighed; efterhånden uden ide om virkeligheden; ude af trit med samfundet; samfundsfjern; manglende sans for detaljer i samfundet; ikke bred indsigt i samfundet

Appendix B3 – Summary Statistics across Professions

Table B3.1: Summary statistics for Free Association Test

Profession	N (respondents)	N (associations)	Mean # of associations	Mean words in associations
Politicians	1275	3895	3,05	1,76
Librarians	174	560	3,22	1,25
Bus drivers	170	362	2,13	1,47
Midwives	169	443	2,62	1,33
Doctors	169	549	3,25	1,37
Police officers	168	487	2,90	1,76
Nurses	163	480	2,94	1,41
Priests	161	546	3,39	1,22
Journalists	160	450	2,81	1,63
Teachers	160	524	3,28	1,44
Farmers	159	528	3,32	1,69
Social workers	155	404	2,61	1,93
Lawyers	154	393	2,55	1,44
Building constructors	153	359	2,35	1,39
Physiotherapists	150	383	2,55	1,36
Bank advisors	146	316	2,16	1,60
Engineers	146	433	2,97	1,35

Appendix B4 – Frequency Table

Table B4.1 depicts the 10 most frequent terms within the two broad subgroups presented in the main article.

Table B4.1: Most Frequent Responses to Cue Word Politicians Across Respondents' Agreement to Political Class Questions

Disagree		Agree	
Word	**Percent**	**Word**	**Percent**
parliament	4.21	untrustworthy	4.00
untrustworthy	2.19	parliament	2.52
power	2.02	selfish	2.13
well paid	1.26	liars	1.92
parties	1.09	career politician	1.50
mette frederiksen	1.04	well paid	1.47
highly educated	0.98	power	1.08
society	0.98	out-of-touch	0.90
debate	0.93	retirement cond.	0.87
elected official	0.87	dishonest	0.84

Note: The table shows the most frequent responses for the cue word politicians for respondents who agree/do not agree that politicians are similar to each other and different from the population. . The percentage is culculated by dividing the frequency of the association within each subgroup by the number of tokens in the group: N = 1829 for the group not agreeing to at least one of the questions. N = 3338 for respondentns agreeing to both questions. The table only shows the 15 most frequent associations made within each group.

Appendix B5 – Fine-grained Political Class Measure

Figure B5.1 shows the relative frequency of words within each of the four subgroups, while Figure B5.2 shows the standardised log-odds ratios comparing the relative frequency of a given word written by respondents who disagree with both statement to the three other subgroups.

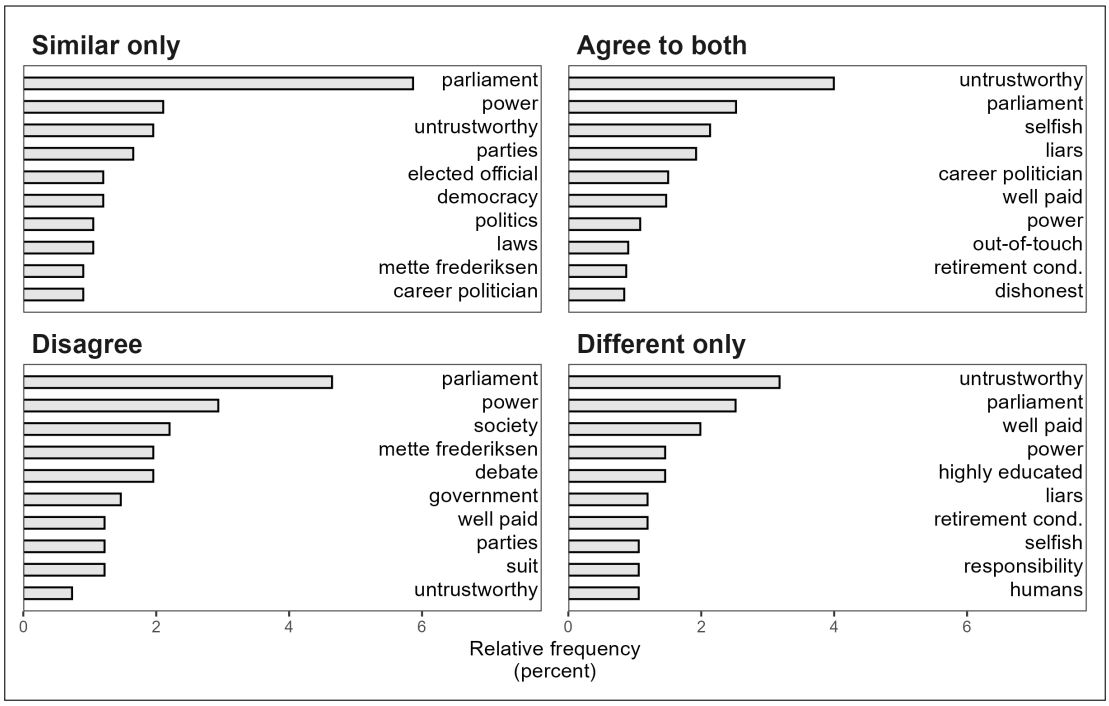


Figure B5.1: Relative Frequency of Associations Across Four Subgroups

Note: The plot shows the relative frequency of associations expressed in percentage.

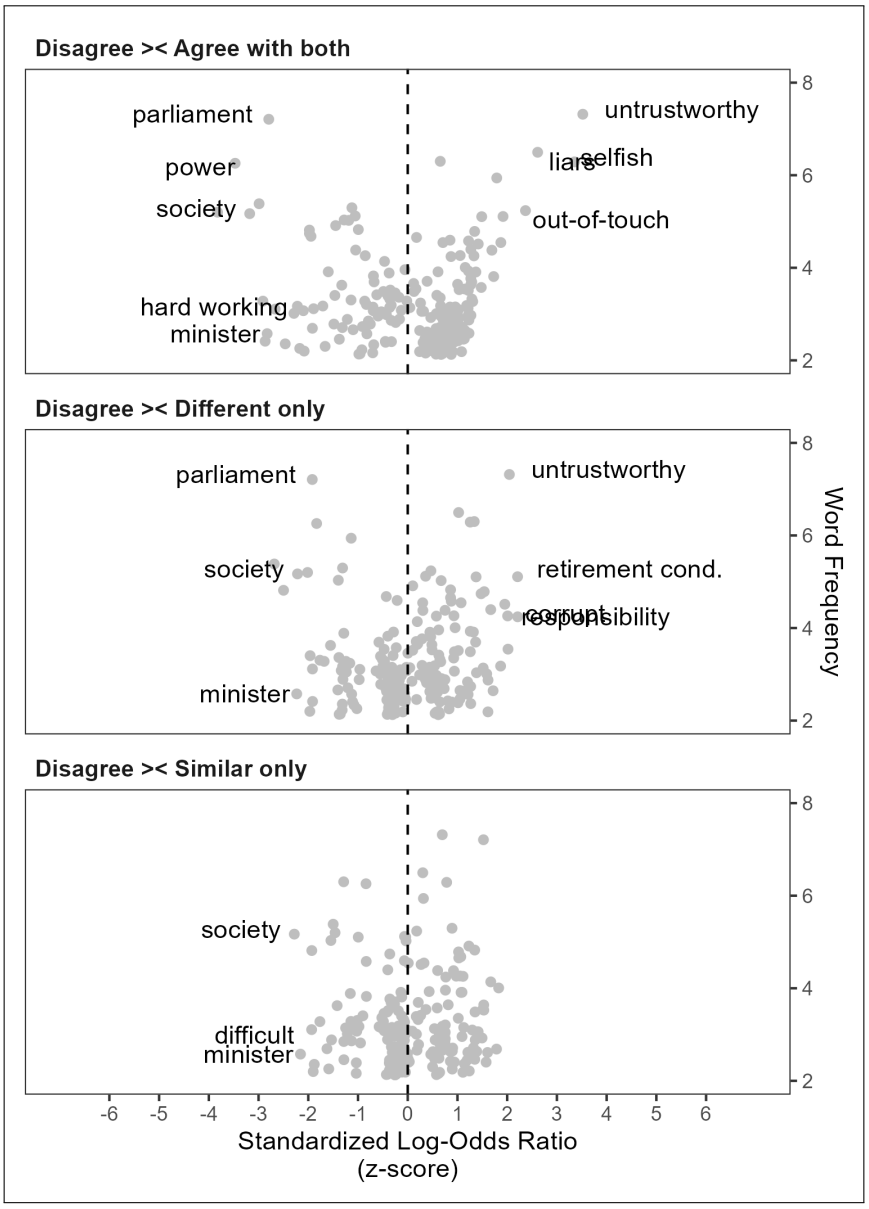


Figure B5.2: Words More Distinctively Used Among the Four Subgroups

Note: The x-axis shows a standardised log-odds ratio comparing respondents within the subgroup for each word occurring more than 5 times in the free association task.

Appendix B6 – Weighted Log-Odds

Figure 3 in the main paper shows how the word usage differs for respondents who agree/disagree with both political class questions. More specifically, the analysis shows an estimated log-odds-ratio for each unique word appearing across the whole corpus of text, where the ratio compares the relative frequency of a given word written by respondents who agree that politicians are similar to each other and different from the population against those who disagree. Following the approach outlined in Monroe, Colaresi, and Quinn (2017), the log-odds-ratio has been standardised (as z-scores) to account for the variance of the estimated ratios. The ratios have further been regularised with a Dirichlet prior to penalise meaningless words (such as "they").

Using the usual bag-of-words assumption (Grimmer and Stewart 2013), I created a grouped document feature matrix counting how many times each unique word appears within each subgroup of respondents. From this, I created a third group containing the total count across all respondents, which is used to set the prior distribution of words (a Dirichlet distribution) that can then be assigned more weight in the estimation of the log–odds–ratios (by setting a scale parameter, α_0). The more weight that is assigned to the prior, the bigger the differences in word usage between the two groups will have to be in order to be detected in the analysis – what is referred to a regularisation or shrinkage.

In the analysis, the scaling parameter is set to a prior sample size of 500. Below I show that the results are similar regardless of the exact value for the scaling parameter, α_0. The Figures show the same results as in Figure 3 in the main article, but with different levels set for the scaling parameter. As is evident when looking at the figures, increasing the scaling parameter shrinks the estimated log–odds–ratios as it should (i.e. the estimates are forced towards 0), yet the relative placement of the words remains very similar.

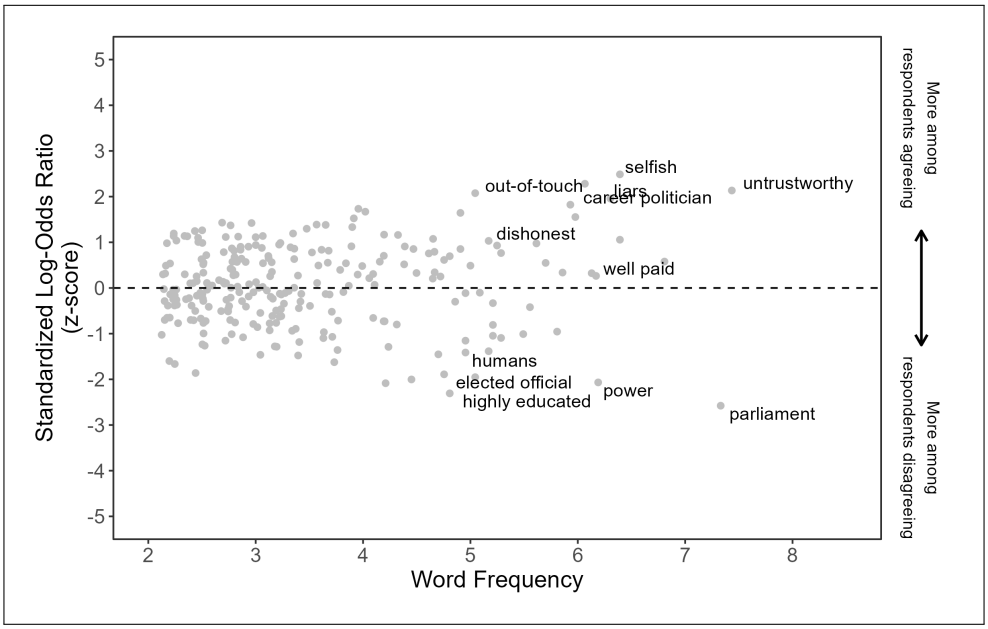


Figure B6.1: Results when scaling parameter is set to 1000

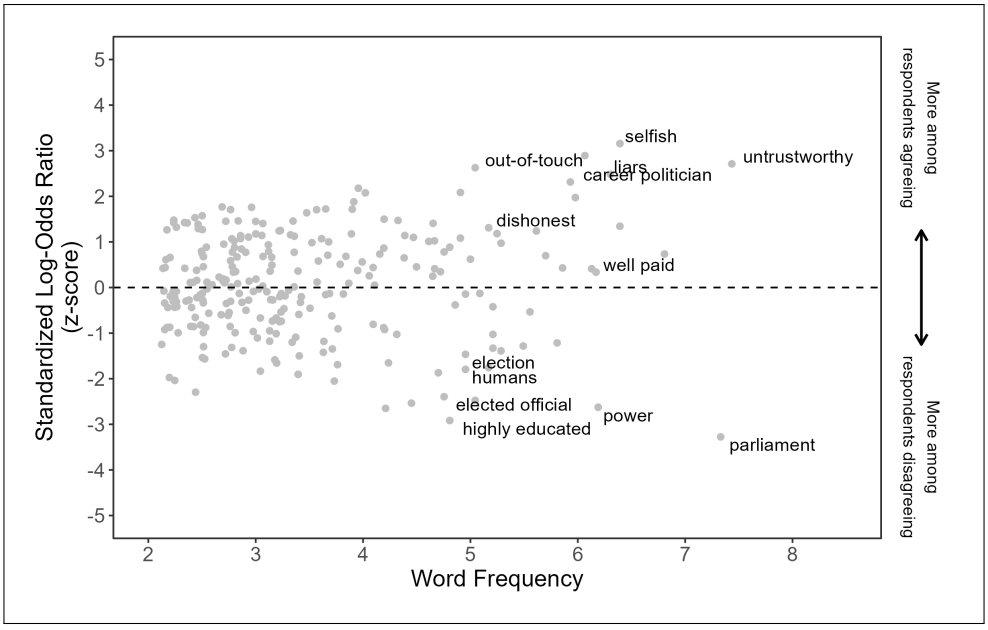


Figure B6.2: Results when scaling parameter is set to 5000

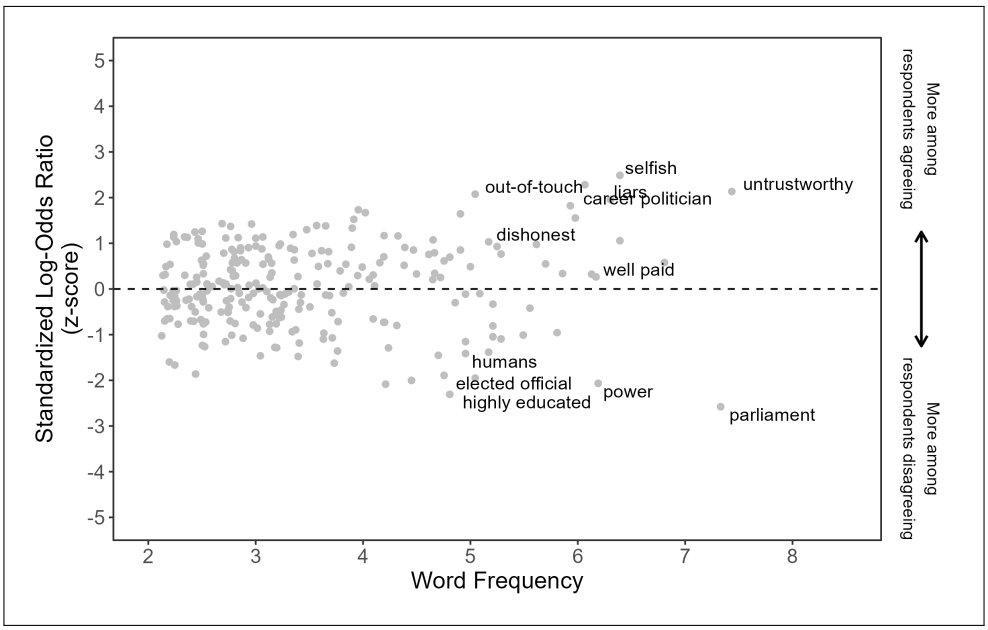


Figure B6.3: Results when scaling parameter is set to 10000

Appendix C – Open-Ended Question

This section contains the appendices relevant to the analysis of the open-ended question. Appendix C1 provides the finalised coding scheme used to manually code the responses.

Appendix C1 – Coding Scheme

The following is the coding scheme for the open-ended question asking respondents to "Describe a typical politician". The unit of analysis is the entire response from a given respondent. Each response is coded on all variables in the coding scheme.

Table C.1: Coding Scheme Overview

Variable	Values / Description
v1: Codability	0) Cannot be further coded 1) Can be further coded
v2: Reason for no code ***if v1 == 0***	1) Don't know 2) Empty/illegible response 3) Refusal of concept 4) Distinguish between politicians
v3: Characteristic	0) No 1) Yes
v3a: What characteristic? ***if yes to v3***	Highly educated (0/1) Upper class (0/1) High income/wealth (0/1) Appearance (0/1), e.g., "suit" or "well dressed" Out of touch (0/1) Career politician (0/1) Lack of work experience (0/1) Sex, race, age (0/1)
v4: Motivation	0) No 1) Yes
v4a: What motivation? ***if yes to v4***	Votes or reelection (0/1) Policy or ideology (0/1) Office or career (0/1) Self gain (0/1) Power (0/1)

Table C.1: Coding Scheme Overview *(continued)*

Variable	**Values / Description**
	Make a change (0/1)
v5: Behavior	0) No 1) Yes
v5a: What behaviour? ***if yes to v5***	Hardworking (0/1) Uncaring (0/1) Spinning and staging (0/1) Talking (0/1) Prevaricating (0/1) Lying (0/1) Break promises (0/1)
v6: Traits	0) No 1) Yes
v6a: What trait? ***if yes to v6***	Dishonest/untrustworthy (0/1) Arrogant (0/1) Selfish (0/1) Extrovert (0/1) Engaged (0/1) Ambitious (0/1)
v7: Abilities	0) No 1) Yes
v7a: What ability? ***if yes to v7***	Articulate (0/1) Knowing (0/1)

Appendix D – Regression tables

This section contains the full output from the regression models used to produce Figures 5 and 6 in the main article.

Table D.1: Regression table for Figure 5

	Political Trust		External efficacy		Democratic satisfaction	
Political class						
Disagree	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Agree	-0.163***	-0.154***	-0.175***	-0.167***	-0.124***	-0.116***
	(0.014)	(0.014)	(0.012)	(0.012)	(0.015)	(0.014)
Gender						
Male	—	Ref.	—	Ref.	—	Ref.
Female	—	0.008	—	-0.007	—	0.000
	—	(0.014)	—	(0.012)	—	(0.015)
Education						
Short	—	Ref.	—	Ref.	—	Ref.
Medium	—	0.007	—	0.018	—	0.031
	—	(0.015)	—	(0.014)	—	(0.017)
Long	—	0.030	—	0.044**	—	0.043*
	—	(0.020)	—	(0.016)	—	(0.020)
Income						
Less than 149.999 kr.	—	Ref.	—	Ref.	—	Ref.
150.000-249.999 kr.	—	0.017	—	0.025	—	0.025
	—	(0.026)	—	(0.024)	—	(0.030)
250.000-349.999 kr.	—	-0.021	—	-0.051*	—	-0.025
	—	(0.026)	—	(0.023)	—	(0.030)
350.000-449.999 kr.	—	-0.010	—	-0.022	—	-0.016
	—	(0.026)	—	(0.023)	—	(0.029)
450.000-599.999 kr.	—	-0.030	—	-0.025	—	-0.041
	—	(0.028)	—	(0.023)	—	(0.030)
600.000-799.999 kr.	—	0.003	—	-0.007	—	0.031
	—	(0.034)	—	(0.029)	—	(0.036)
Above 800.000 kr.	—	-0.050	—	-0.051	—	-0.068
	—	(0.044)	—	(0.035)	—	(0.051)
DK/Don't want to answer	—	-0.044	—	-0.059*	—	-0.059
	—	(0.027)	—	(0.024)	—	(0.031)
Subjective class						
Working class	—	Ref.	—	Ref.	—	Ref.

Middle class	—	0.081***	—	0.063***	—	0.073***
	—	(0.016)	—	(0.015)	—	(0.018)
Upper class	—	0.094***	—	0.074***	—	0.077**
	—	(0.022)	—	(0.020)	—	(0.025)
Constant	0.476***	0.335***	0.466***	0.374***	0.648***	0.553***
	(0.010)	(0.027)	(0.010)	(0.023)	(0.011)	(0.030)
Num.Obs.	1099	1099	1099	1099	1099	1099
R2	0.114	0.159	0.159	0.203	0.061	0.099

Robust standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table D.2: Regression table for Figure 6

	Political Trust		External efficacy		Democratic satisfaction	
Political class						
Disagree to both	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Similar only	0.017	0.014	-0.044	-0.050*	0.026	0.025
	(0.027)	(0.026)	(0.024)	(0.024)	(0.026)	(0.026)
Different only	-0.128***	-0.127***	-0.174***	-0.176***	-0.113***	-0.111***
	(0.028)	(0.027)	(0.024)	(0.023)	(0.028)	(0.028)
Agree to both	-0.208***	-0.200***	-0.261***	-0.257***	-0.159***	-0.152***
	(0.024)	(0.024)	(0.021)	(0.020)	(0.023)	(0.023)
Gender						
Male	—	Ref.	—	Ref.	—	Ref.
Female	—	0.010	—	-0.007	—	0.002
	—	(0.013)	—	(0.012)	—	(0.014)
Education						
Short	—	Ref.	—	Ref.	—	Ref.
Medium	—	0.011	—	0.021	—	0.035*
	—	(0.015)	—	(0.013)	—	(0.017)
Long	—	0.037	—	0.051**	—	0.050*
	—	(0.019)	—	(0.016)	—	(0.020)
Income						
Less than 149.999 kr.	—	Ref.	—	Ref.	—	Ref.
150.000-249.999 kr.	—	0.023	—	0.027	—	0.031
	—	(0.026)	—	(0.023)	—	(0.030)
250.000-349.999 kr.	—	-0.014	—	-0.047*	—	-0.017
	—	(0.026)	—	(0.023)	—	(0.030)
350.000-449.999 kr.	—	0.000	—	-0.014	—	-0.005
	—	(0.026)	—	(0.022)	—	(0.029)
450.000-599.999 kr.	—	-0.024	—	-0.022	—	-0.035
	—	(0.028)	—	(0.023)	—	(0.030)
600.000-799.999 kr.	—	0.009	—	-0.005	—	0.037
	—	(0.033)	—	(0.028)	—	(0.035)
Above 800.000 kr.	—	-0.035	—	-0.039	—	-0.053
	—	(0.043)	—	(0.035)	—	(0.050)
Don't know/Don't want to answer	—	-0.037	—	-0.056*	—	-0.053

	—	(0.027)	—	(0.024)	—	(0.031)
Subjective class						
Working class	—	Ref.	—	Ref.	—	Ref.
Middle class	—	0.075***	—	0.057***	—	0.067***
	—	(0.015)	—	(0.014)	—	(0.018)
Upper class	—	0.081***	—	0.061**	—	0.065**
	—	(0.022)	—	(0.020)	—	(0.024)
Constant	0.520***	0.377***	0.552***	0.461***	0.683***	0.583***
	(0.022)	(0.033)	(0.019)	(0.028)	(0.021)	(0.035)
Num.Obs.	1099	1099	1099	1099	1099	1099
R2	0.148	0.190	0.206	0.249	0.088	0.125

Robust standard errors in parentheses
* p < 0.05, ** p < 0.01, *** p < 0.001

Appendix E – Ethical Considerations

The research presented in the manuscript complies with APSA's Principles and Guidance for Human Subjects Research. The data is collected in Denmark, and the data collection process complies with relevant Danish legislation. The study is exempt from formal ethics review in Denmark following Section 14 (2) of the act underlying the Danish National Research Ethics Committee, specifying that "notification of questionnaire surveys ... to the system of research ethics committee system is only required if the project involves human biological material."

The human subjects were a sample of Danish Adults (18 years or older) and did not involve vulnerable individuals or groups, such as minors, patients, or persons unable to give informed consent. All study participants are volunteers recruited to ensure diversity and representativeness on key demographics (i.e., gender, age, and geography). Data was collected by Norstat A/S from the company's standing panel of respondents. This means that all respondents will have signed up to receive surveys of various kinds and be accustomed to responding to surveys as well as to Norstat's procedures for respondent remuneration.

As part of the introduction to the surveys, respondents were provided with information about the general aim of the research, thereby laying the ground for obtaining their informed consent. Likewise, respondents were informed that their responses would be anonymous.

In sum, I believe the research study is in full compliance with current guidelines and principles for ethical and transparent research in our field.